

Day 07 - SpaceAgent YouTube 100% Closure

Date: 2026-05-10

Branch: to-knowledge-mc

Lane: SpaceAgent YouTube

Status: GO for developer-side YouTube metadata/transcript connector proof

Blocker class: NONE

Objective

Close the SpaceAgent YouTube lane by proving the current runtime can inspect a public YouTube URL through the approved read-only metadata/transcript path.

Safety

- No '.env' changes.
- No secrets printed.
- No auth weakening.
- No public local exposure added.
- No full video download.
- No login, age, region, copyright, or paywall bypass.
- No external writes.
- SpaceAgent remains a read-only research specialist.

Implementation

Fixed the live transcript adapter for the installed 'youtube-transcript-api' runtime package.

Previous connector behavior:

- Dependency check could become READY after install.
- Live transcript lookup still failed because the connector used the old 'YouTubeTranscriptApi.get_transcript(...)' API.

New connector behavior:

- Uses 'YouTubeTranscriptApi().fetch(video_id)'.
- Reads transcript snippet fields from the current object shape.
- Keeps bounded transcript extraction to the first 30 snippets.
- Keeps full video download blocked by default.

Files Changed

- 'src/lib/space-agent-youtube-connector.ts'
- 'runtime/day-07-spaceagent-youtube-closure.md'
- 'runtime/day-07-spaceagent-youtube-closure.pdf'

Runtime Dependency

The current host runtime has 'youtube-transcript-api' installed for 'python3'.

Runtime dependency proof:

- Python package: 'youtube-transcript-api'
- Import check: passed
- Connector status route: READY

Rollback note for runtime dependency:

- If this lane is intentionally reverted, the code rollback is 'git revert <day07_commit_sha>'.
- If the owner also wants to remove the host Python package, run a separate owner/admin action: 'python3 -m pip uninstall youtube-transcript-api'.

Routes Proved

Authenticated with runtime API key context:

GET '/api/gateway/space-agent/youtube/status'

Result:

- HTTP 200
- 'status': 'ready_for_transcript_smoke'
- 'canonical_status': 'READY'
- 'blocker_class': 'NONE'
- 'blocked_reason': 'null'
- 'transcript_connector_proven': 'true'
- 'metadata_enabled': 'true'

POST '/api/gateway/space-agent/youtube/research'

Input:

- Public video URL: 'https://www.youtube.com/watch?v=dQw4w9WgXcQ'
- Responsible agent: 'agent_zero'

Result:

- HTTP 200
- 'ok': 'true'
- 'blocked_reason': 'null'
- 'packet_status': 'ready'
- 'transcript_status': 'available'
- 'captions_available': 'true'
- 'transcript_segments': 30
- 'key_claims': 12
- 'title': Rick Astley - Never Gonna Give You Up (Official Video) (4K Remaster)
- 'channel': Rick Astley
- 'publish_date': unavailable from the oEmbed metadata path
- 'full_video_download_allowed': 'false'
- 'no_secrets_exposed': 'true'
- 'no_raw_paths': 'true'

First transcript proof snippet:

- Start: 1.36s
- End: 3.04s
- Text: '[music marker]'
- Source: official transcript

Tests And Checks

Passed:

- 'git diff --check'
- 'pnpm exec vitest run src/lib/space-agent-youtube-connector.test.ts src/lib/space-agent-youtube-scenarios.test.ts src/lib/space-agent-research.test.ts'
- 'pnpm run typecheck'
- 'pnpm run build'
- 'pnpm test'
- 'node scripts/check-mission-control-route-rendering.mjs http://127.0.0.1:3337 /api/gateway/space-agent/youtube/status /api/gateway/space-agent/youtube/research'
- 'node scripts/check-protected-file-invariants.mjs'

Test totals:

- Full suite: 171 files / 1356 tests passed.
- Targeted YouTube suite: 3 files / 35 tests passed.

Production Runtime

Local production runtime restarted from the rebuilt standalone bundle.

Runtime:

- Bind: '127.0.0.1:3337'
- PID after restart: 23658

- '/login': included in route-rendering smoke and returned 200.
- No new public exposure.

Audit / Proof Packet

Proof route:

- '/api/gateway/space-agent/youtube/status'

Research route:

- '/api/gateway/space-agent/youtube/research'

Audit result:

- Read-only proof only.
- No Bridge Session required because no external write occurred.
- No audit row written for execution because no execution/write was dispatched.

Remaining Blockers

None for developer-side YouTube metadata/transcript proof.

External limitations still apply per policy:

- Private, login-required, age-gated, region-locked, unavailable, or transcript-disabled videos remain blocked or limited.
- Full video download remains blocked by default.

Rollback

Code rollback:

```
""bash
git revert <day07_commit_sha>
""
```

Optional host dependency cleanup, only if owner/admin wants to remove the runtime package:

```
""bash
python3 -m pip uninstall youtube-transcript-api
""
```

Closeout

Day 07 SpaceAgent YouTube is closed for developer-side acceptance.

Next day automatically started:

- Day 08 - SpaceAgent Firecrawl 100% Closure